



UNIVERSITI TEKNIKAL MALAYSIA MELAKA
FACULTY OF INFORMATION AND COMMUNICATION
TECHNOLOGY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | TAJUK PROJEK YANG DICADANGKAN

AN INTELLIGENT MANAGEMENT FRAMEWORK FOR MITIGATING SOCIAL-ENGINEERED SPAM AND CYBER THREATS IN UNIVERSITY SOCIAL GROUPS

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C PROJECT INFORMATION | MAKLUMAT PROJEK

(i) Executive Summary of Project Proposal [Maximum 300 words]

As university social groups example like WhatsApp and Telegram become indispensable for campus communication, they have simultaneously emerged as primary targets for social engineering attacks and malicious spam. This project focuses on developing an intelligent management framework to reduce the impact of social-engineered spam and cyber threats in university social groups as current administrative practices in these groups rely heavily on manual moderation which is quite inefficient and often fails to detect localized multi-lingual slangs. This "trust-based" manual monitoring model creates a significant security gap, allowing phishing links and fraudulent information to spread rapidly before any intervention occurs.

To address these challenges, an Intelligent Management Framework will be design to proactively mitigate social-engineered threats in this project. The core innovation lies in the integration of Random Forest machine learning algorithms tailored for the Malaysian university linguistic context. The framework aims to automate the identification of threat patterns and map them to a structured administrative response system unlike the most traditional filter.

The main deliverable of this project is a lightweight demonstration prototype (proof of concept) based on Streamlit. This tool will enhance the capabilities of group administrators by providing real-time threat classification and automated mitigation recommendations, such as user warnings or ban suggestions. By transforming a manual oversight model into an intelligent, proactive management model, this project provides a robust solution for ensuring the long-term security and integrity of the campus social network environment.

(ii) Detailed Proposal of the Project

(a) Introduction (Project Background and Problem Statements)

The digital transformation of academic communication has made instant messaging platforms such as WhatsApp and Telegram which is a primary infrastructure for student interaction at local university in Malaysia. These platforms are key hubs for disseminating information such as scholarships, academic information, and departmental announcements. However, this high degree of interconnectivity has also brought about a sophisticated attack avenue that is human-centric social engineering attacks. Unlike traditional cyber-level threats, social engineering attacks exploit the "trust dependency" within campus groups, with malicious attackers impersonating university officials or student representatives.

A prime example of this growing threat is the emergence of fraudulent emails and messages targeting UTeM student webmail accounts, specifically regarding "unpaid tuition fees" or "urgent administrative fines." These deceptive communications often mirror the official formatting of university bursary departments to create a false sense of urgency. When such tactics are transitioned into social media groups, the risk is amplified by the perceived intimacy of the platform. Recent cybersecurity trends in Malaysian higher education indicate a rise in cases of localized phishing attacks that exploit the unique language habits of student groups. These attacks are no longer limited to standard English but often use "Manglish" and multilingual code switching to masquerade as legitimate information, bypassing common security filters. Therefore, current security measures, heavily reliant on manual intervention by group administrators, are increasingly struggling to cope with the sheer number and speed of modern deception techniques.

(b) Problem Statement

1. **Operational Bottlenecks:** The 24/7 nature of social media makes manual moderation by student or faculty administrators nearly impossible and it will be costly if it is done by manual, resulting in excessively high response delays during periods of high phishing activity.
2. **Language Circumvention:** Malicious actors frequently use local slang and code-switching (Malay-English hybrids) to avoid being detected by standard English-centric keyword filters.
3. **Subjective Decision-Making:** The lack of a standardized framework leads to inconsistent management practices, with different moderators potentially handling similar threats differently.

(c) Methodology Integration

To ensure a structured development, this research follows a four-stage experimental workflow:

1. **Data Acquisition:** Collecting and anonymizing text samples of common campus scams, including "fake tuition" and "scholarship phishing" texts.
2. **Model Training:** Using Text Feature Extraction to train Random Forest and SVM models to recognize the intent behind deceptive messages.
3. **Framework Design:** Creating an administrative response matrix that links detected threat levels to specific actions like "Alert Admin" or "Restrict User."
4. **Prototype Development:** Building a user-friendly dashboard with Streamlit to showcase how these security decisions can be automated.

(d) Objectives of the Project

This project embarks on the following objectives:

1. To compile and pre-process a localized dataset of social-engineered text incidents targeting the Malaysian higher education sector.
2. To benchmark the detection accuracy of Random Forest versus SVM in identifying threats within multilingual social media contexts.
3. To construct a proactive management framework that provides objective, risk-based response protocols for group administrators.
4. To implement a web-based prototype dashboard to validate the feasibility of the proposed automated mitigation workflow.

	(c) Scope of the Project The project boundaries are defined as follows: 1. Data Domain and Platform: This research primarily targets text-based social engineering threats, such as phishing links, fraudulent tuition payment notifications, and scholarship scams, that are propagated in WhatsApp and Telegram social groups at Malaysian universities. 2. Language Focus: The detection model is specifically designed for localized language patterns, including Malay-English mixed languages (Manglish) and Malay-English code-switching, to overcome the limitations of standard English-centric security filters. 3. Technical Implementation: Machine learning was performed using Python during development, and the front-end management panel was built using Streamlit. Core algorithms were limited to supervised learning models, specifically random forests and support vector machines (SVMs), for benchmarking. 4. Management Framework: The framework is designed with a focus on proactive decision support logic like risk scoring and response recommendations rather than entirely bot-based autonomous auditing. 5. Exclusions: This project does not cover image-based threat detection (OCR), deep packet inspection of encrypted traffic, or hardware-level security protocols.
	(d) Expected Outcome/ Proposed Solution Upon successful completion, this project will deliver the following outcomes: 1. Intelligent Threat Detection Engine: A validated machine learning model capable of identifying social engineering intent in localized campus communications with high accuracy and a low false positive rate. 2. Proactive Management Framework: A structured management response matrix providing group administrators with standardized, risk-based mitigation options such as "warning," "mute," or "ban" suggestions to eliminate subjective decision-making. 3. Functional Web Prototype: A lightweight, user-friendly Streamlit-based dashboard demonstrating real-time message classification, risk scoring, and automated mitigation suggestions for university social group administrators. 4. Performance Comparison Analysis: A comprehensive technical report documenting benchmark results between Random Forest and Support Vector Machine (SVM), providing data-driven justification for the final model selection. 5. Localized Security Dataset: A curated and anonymized social engineering attack dataset specifically tailored to the Malaysian higher education population, serving as a resource for future campus security research.

D REFERENCES | RUJUKAN

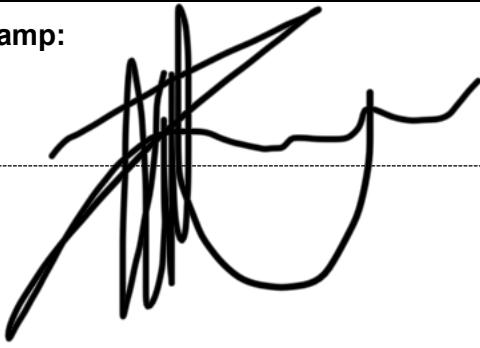
State your references (Minimum 10 references)

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E DECLARATION BY STUDENT AKUAN PELAJAR	
(i)	Date: Student's Signature: _____
E RECOMMENDED BY SUPERVISOR PERAKUAN OLEH PENYELIA	
(ii)	Recommended ☒ Not Recommended ☐
	Comments:
	Supervisor's Name:
MEGAT MUHAMMAD RIDZUAN BIN MEGAT KHAIRULAZHAR	

Signature & Stamp:

A handwritten signature in black ink, consisting of several loops and a long horizontal stroke, positioned above a dashed horizontal line.

Date: 13/3/2026